BESS Technical Gate Review

Pre-Investment Screening — Utility-Scale BESS

Client:	Public Reference Case — Anonymized
Project:	Utility-Scale BESS (200 MW / 800 MWh) Southwest Desert, United States
Report Number:	BiP-SAMPLE-2025-SW
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Site Coordinates:	Anonymized — Southwest Desert, USA

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Important Notices

Scope of This Document

This document is a **public reference case** produced by **BiP Omega** to illustrate the structure and analytical depth of a Pre-Investment Technical Assessment mandate. All project identifiers, geographic coordinates, counterparty names, and commercially sensitive parameters have been anonymized. Findings are derived from the documentation package made available for this assessment; no site visit was conducted as part of this mandate scope.

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Executive Summary

TECHNICAL GATE VERDICT

CONDITIONAL NO-GO

Advance to full technical due diligence only after resolving the three critical evidence gaps identified in this report.

Bankability Index:	31.2 / 100	*(Institutional threshold: 60.0)*
Critical Silences:	**3 CRITICAL, 1 HIGH**	
Unquantified Exposure:	**$10–20M+ annually**	

This report presents the findings of a **Pre-Investment Technical Assessment** conducted by BiP Omega on a 200 MW / 800 MWh utility-scale BESS project located in a high-temperature desert environment in the U.S. Southwest. The assessment was conducted under a *pre-financial-close investment committee mandate*, with the objective of determining whether the available technical documentation is sufficient to support a GO recommendation to full diligence.

The analysis identified three categories of material evidence gap that currently prevent a confident bankability assessment:

Critical Silence	Financial Consequence	Verdict Impact
Grid Interconnection Status	$8–15M financing cost exposure from delay risk	CRITICAL
Thermal Management Architecture	$2–5M annual revenue at risk during peak windows	CRITICAL
Warranty & Performance Guarantees	DSCR covenant breach if performance < projections	CRITICAL
Fire Safety Certification (UL9540A)	150–300% insurance premium increase	HIGH

Table 1: Summary of Material Evidence Gaps and Financial Consequences

Mandate scope. This assessment operated under a pre-investment mandate with documentation provided by the project sponsor. It is classified as a *Preliminary Assessment* — it establishes which technical questions must be answered before capital can be committed, and provides a structured set of direct questions for the next stage of diligence. A full *Lender's Technical Review* or *Acquisition Due Diligence* mandate would require access to manufacturer test data, the data room, and counterparty contractual documentation.

Chapter 1

Evidence Classification Register

Every parameter in this assessment is classified according to its evidential basis. No value is assumed without disclosure. No unknown is substituted with a default.

Parameter	Value	Confidence	Source
Power Capacity	200 MW	**VERIFIED**	Sponsor DD Package
Energy Capacity	800 MWh (4h)	**VERIFIED**	Sponsor DD Package
Battery Chemistry	Li-ion (LFP)	**VERIFIED**	Sponsor DD Package
Site Area	58 acres	**VERIFIED**	Sponsor DD Package
Max Ambient Temperature	47°C	**VERIFIED**	Sponsor DD Package
Peak Price Window	4:00–7:00 PM	**VERIFIED**	ISO Market Data
HVAC Parasitic Load Range	3–7%	**VERIFIED**	Sponsor DD Package
Gen-Tie Length	18.3 km	**VERIFIED**	Permit Filing
Interconnection Queue Position	*Filed*	**ASSUMED**	Utility Filing (date unconfirmed)
Estimated COD	Q3 2027	**ASSUMED**	Sponsor Projection
Degradation Curve (Year 10)	~80%	**ASSUMED**	Manufacturer datasheet
Revenue (CAISO RA + DLAP)	$18–24M / yr	**ASSUMED**	Derived from capacity; not confirmed
Thermal Derating Curves	*Unknown*	**CRITICAL SILENCE**	Not disclosed
Integrator / BMS Vendor	*Unknown*	**CRITICAL SILENCE**	Not disclosed
Warranty Terms (throughput, EOL)	*Unknown*	**CRITICAL SILENCE**	Not disclosed
UL9540A Test Report	*Unknown*	**CRITICAL SILENCE**	Not disclosed
Interconnection Agreement Status	*Unknown*	**CRITICAL SILENCE**	Utility confirmation not available

Table 1.1: Evidence-Grade Parameter Register — Full Classification

Classification key. **VERIFIED** — confirmed from source documentation. **ASSUMED** — derived with disclosed basis; carries inherent uncertainty. **CRITICAL SILENCE** — not disclosed and material to the investment or financing decision; cannot be substituted with assumption without misrepresenting confidence.

Chapter 2

Critical Silence Analysis

The following material evidence gaps were identified during this assessment. Each gap is documented with its category, what was disclosed, what is missing, the specific financial or technical consequence of the absence, and the exact information required to resolve it.

2.1 CS-001 — Grid Interconnection Status

- **Category:** Regulatory / Commercial
- **Disclosed:** Interconnection application filed with the host utility. Queue position and study phase unconfirmed.
- **Missing:** Executed Interconnection Agreement or queue study results confirming position, upgrade obligations, and cost allocation.
- **Financial consequence:** Projects in active Phase 2 restudy face 12–24-month delays and upgrade costs of $8–15M from revised network upgrade allocations. This directly impacts the financing timeline and carrying cost structure.
- **Verdict impact:** COD assumption of Q3 2027 is undefendable without an executed agreement or a Phase 2 study with cost certainty.
- **Direct question:** *"Provide the current interconnection queue position, study phase, and any network upgrade cost estimates or cost cap provisions from the utility."*

2.2 CS-002 — Thermal Management Architecture

- **Category:** Technical / Performance
- **Disclosed:** Maximum ambient temperature of 47°C confirmed. HVAC parasitic load range of 3–7% disclosed.
- **Missing:** BMS thermal derating curves from 30°C to 50°C; HVAC cooling capacity (kW) and N+1 redundancy configuration; thermal setpoints triggering output curtailment.
- **Financial consequence:** LFP cells operating above nominal thermal envelope derate measurably during the 4:00–7:00 PM peak window — precisely when ISO energy prices and RA obligations peak. Estimated revenue exposure: $2–5M annually if derating is not modeled and contractually protected.
- **Verdict impact:** Revenue projections are unverifiable. Any bankability model built on capacity without thermal validation overstates the denominator of the DSCR calculation.
- **Direct question:** *"Provide BMS thermal derating curves from 30°C to 50°C and HVAC cooling capacity specifications with redundancy configuration."*

2.3 CS-003 — Warranty and Performance Guarantees

- **Category:** Contractual / Commercial
- **Disclosed:** Power capacity (200 MW) specified. Battery chemistry confirmed.
- **Missing:** Throughput warranty (MWh over project life), availability guarantee (%), end-of-life (EOL) capacity floor, and degradation curve terms.
- **Financial consequence:** Without bankable warranty terms, there is no contractual recourse if actual performance deviates from projections. A degradation shortfall of 5% over 10 years against DSCR projections is sufficient to trigger covenant breach in standard project finance structures.
- **Verdict impact:** Lenders cannot size debt against unwarrantied performance. This alone is a lender's independent engineer blocking condition.
- **Direct question:** *"Provide the executed or draft warranty agreement covering throughput, availability, EOL capacity, and degradation curve with degradation guarantee floor."*

2.4 CS-004 — Fire Safety Certification (UL9540A)

- **Category:** Safety / Insurance
- **Disclosed:** LFP chemistry specified. No fire propagation or suppression documentation provided.
- **Missing:** UL9540A cell-level and module-level test reports; fire suppression system design; AHJ (Authority Having Jurisdiction) approval status.
- **Financial consequence:** Insurers increasingly require UL9540A certification as a condition of coverage. Without it, premiums increase 150–300% above standard rates, or coverage is excluded for thermal runaway events — the primary insurance risk category for LFP utility-scale projects.
- **Verdict impact:** Rated HIGH (not CRITICAL) because LFP chemistry has a lower thermal runaway propagation risk than NMC. However, the absence of documentation creates an unresolved insurance underwriting question that must be resolved before construction financing is placed.
- **Direct question:** *"Provide UL9540A test reports at cell and module level, fire suppression system design specifications, and AHJ approval correspondence."*

Chapter 3

Evidence-Based Risk Register

The risk register below covers only risks for which documentary evidence exists to support the trigger condition. Risks without an evidential basis are excluded. Probability and impact are scored 1–5; color coding reflects the composite risk score.

Risk	P	I	Score	Basis
Interconnection delay (COD slip)	4	5	**20**	Queue position unconfirmed; Phase 2 restudy risk
Thermal derating at peak	4	5	**20**	47°C confirmed; derating curves absent
Warranty gap / DSCR breach	4	5	**20**	No throughput or EOL warranty disclosed
Insurance exclusion / premium spike	4	4	**16**	UL9540A documentation absent
HVAC parasitic load overrun	3	4	**12**	3–7% range disclosed; no capacity confirmation
Civil infrastructure (gen-tie ROW)	3	3	**9**	18.3 km gen-tie crosses federal land; ROW unconfirmed

Table 3.1: Evidence-Based Risk Register (P = Probability 1–5, I = Impact 1–5)

Risk score color key: Score > 15 = **Critical** (red)   Score 10–15 = **High** (amber)   Score ≤ 9 = **Medium/Low** (green)

Chapter 4

Financial Exposure Analysis

The table below identifies quantifiable annual exposure ranges attributable to the evidence gaps in this report. These figures represent the delta between the sponsor's undiscounted revenue assumptions and the technically defensible range given available documentation.

Risk Category	Basis	Annual Exposure
Thermal derating at peak	Revenue loss during 4:00–7:00 PM window Jul–Sep	**$2–5M**
Interconnection delay (financing cost)	12–24 month COD slip; carrying cost at 6.5%	**$8–15M one-time**
HVAC parasitic overrun	1–2% additional parasitic above disclosed range	**$0.5–1.5M**
Insurance premium increase (UL9540A absent)	150–300% above standard rate	**$1.5–4M**
DSCR covenant breach (warranty gap)	Performance shortfall vs. projection	*Not quantifiable without warranty terms*
Total Identified Exposure		**$12–26M+**

Table 4.1: Financial Risk Exposure Summary

[!] CRITICAL WARNING

These figures represent **unquantified exposure**, not projected losses. The actual impact depends on terms that have not been disclosed. Investment decisions made without resolving these gaps carry risk that cannot currently be bounded.

Chapter 5

Technical Gate Decision

5.1 Current CONDITIONAL NO-GO Conditions

Current Status — CONDITIONAL NO-GO

- × Interconnection agreement not executed; COD unconfirmable
- × Thermal derating curves absent; peak revenue unverifiable
- × Warranty and performance guarantee terms not disclosed
- × UL9540A certification documentation not provided (HIGH)
- × Bankability index below institutional threshold (31.2 / 100)

5.2 Conditions Required for GO Clearance

Required for GO Recommendation

- ✓ Executed Interconnection Agreement *or* binding Phase 2 study with capped upgrade obligations
- ✓ BMS thermal derating curves validated against 47°C site conditions and HVAC capacity confirmed
- ✓ Executed warranty agreement with bankable throughput, availability, and EOL capacity guarantees reviewed
- ✓ UL9540A certification confirmed and insurance underwriting terms obtained

Disposition. The project exhibits sufficient scale (200 MW / 800 MWh), acceptable chemistry (LFP), and a plausible site configuration to merit continued diligence. The CONDITIONAL NO-GO is not a rejection of the project — it is a documented statement that the current evidence base does not support a defensible GO recommendation. The investment committee cannot be protected by a GO verdict it cannot defend.

Chapter 6

Suggested Areas for Further Technical Review

A **Lender's Technical Review** mandate would extend this assessment with the following scope additions:

Scope Item	What It Resolves
Interconnection agreement review	COD certainty, upgrade cost obligations, curtailment provisions
Manufacturer BMS validation	Thermal derating curves, capacity under site conditions
Warranty structure analysis	DSCR defensibility, lender step-in rights, EOL floor
UL9540A and fire suppression review	Insurance underwriting basis, AHJ approval path
Revenue model stress test	Degradation-adjusted RA and DLAP revenue under P90 conditions

Table 6.1: Lender's Technical Review — Scope Summary

A full Lender's Technical Review mandate operates on the same analytical framework as this preliminary assessment — the same evidence classification, the same risk documentation standard, the same traceable decision logic. What changes is the scope of evidence ingested and the deliverable format required by the financing institution.

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